

CHAPTER 5: MAGNETISM & ELECTROMAGNETIC INDUCTION

At the end of this chapter, you'll be able to:

SUBTOPIC	LEARNING OUTCOMES						
5.1	Magnetic field a) Define magnetic field. b) Identify magnetic field sources. *e.g. bar magnet and current-carrying conductor (straight wire and circular coil), and Earth magnetic field.						
5.2	Magnetic field produced by current-carrying conductor a) Sketch magnetic field lines patterns of current carrying straight wire, circular coil and solenoid. b) Use the magnetic field <table><tr><td>for a long straight wire:</td><td>at the center of a circular coil:</td><td>at the center of solenoid:</td></tr><tr><td>$B = \frac{\mu_0 I}{2\pi r}$</td><td>$B = \frac{\mu_0 I}{2r}$</td><td>$B = \mu_0 nI$</td></tr></table>	for a long straight wire:	at the center of a circular coil:	at the center of solenoid:	$B = \frac{\mu_0 I}{2\pi r}$	$B = \frac{\mu_0 I}{2r}$	$B = \mu_0 nI$
for a long straight wire:	at the center of a circular coil:	at the center of solenoid:					
$B = \frac{\mu_0 I}{2\pi r}$	$B = \frac{\mu_0 I}{2r}$	$B = \mu_0 nI$					
5.3	Magnetic flux a) Define and use magnetic flux, $\phi = \vec{B} \cdot \vec{A}$ b) Use magnetic flux linkage, $\Phi = N\phi$						
5.4	Induced emf a) State and use Faraday’s law, $\varepsilon = -\frac{d\phi}{dt}$ b) State and apply Lenz’s law to find the direction of induced current. e) Use induced emf in <table><tr><td>a straight conductor:</td><td>a coil:</td></tr><tr><td>$\varepsilon = Blv \sin\theta$</td><td>i. $\varepsilon = -NA\frac{dB}{dt}$</td></tr><tr><td></td><td>ii. $\varepsilon = -NB\frac{dA}{dt}$</td></tr></table>	a straight conductor:	a coil:	$\varepsilon = Blv \sin\theta$	i. $\varepsilon = -NA\frac{dB}{dt}$		ii. $\varepsilon = -NB\frac{dA}{dt}$
a straight conductor:	a coil:						
$\varepsilon = Blv \sin\theta$	i. $\varepsilon = -NA\frac{dB}{dt}$						
	ii. $\varepsilon = -NB\frac{dA}{dt}$						

5.1 MAGNETISM

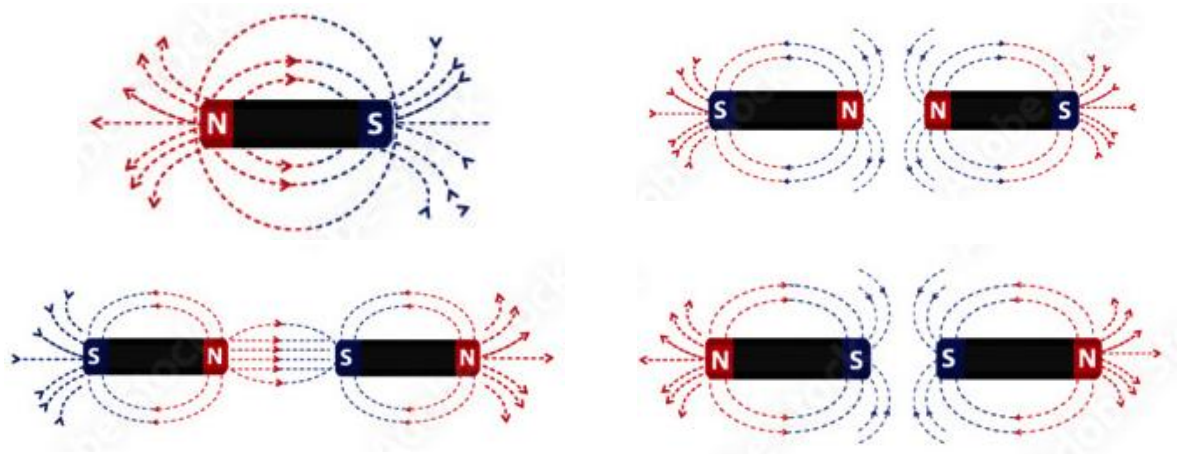
Magnetic Field

1. Magnetic fields defined as a **region around a magnet** where a **magnetic force can be experienced**.
2. Is a **vector** quantity.
3. Magnetic field has two poles, called **north (N)** and **south (S)**.

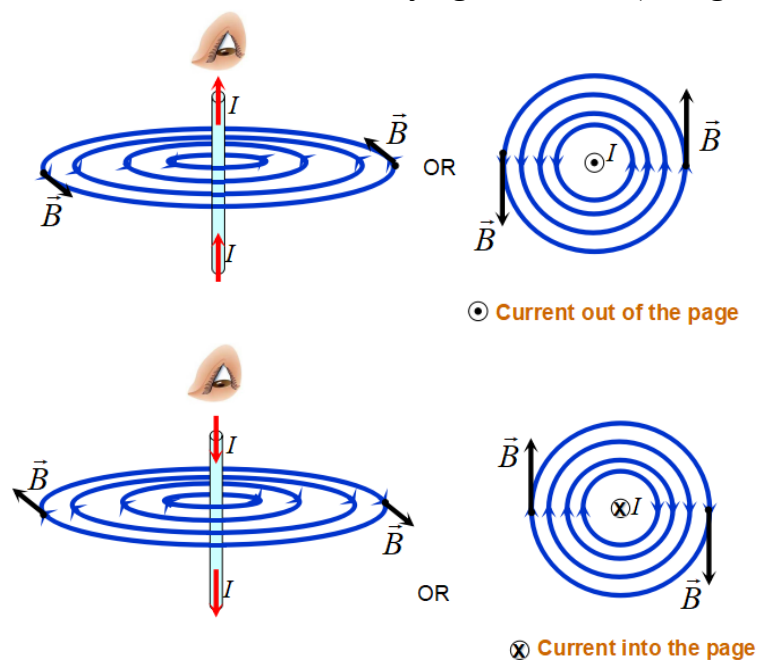
Magnetic Field Sources

1. **Earth** – natural permanent magnet.
2. **Bar magnet** – permanent magnet with a typical magnetic field.
3. A **moving charge** or a **current carrying conductor** creates a magnetic field in the surrounding space.

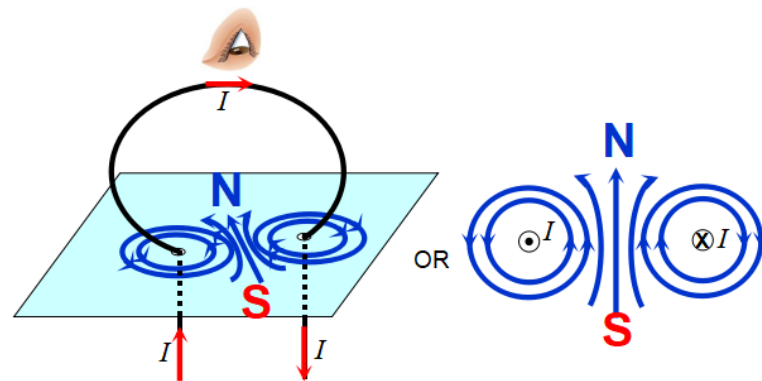
Magnetic Field Lines of Bar Magnet



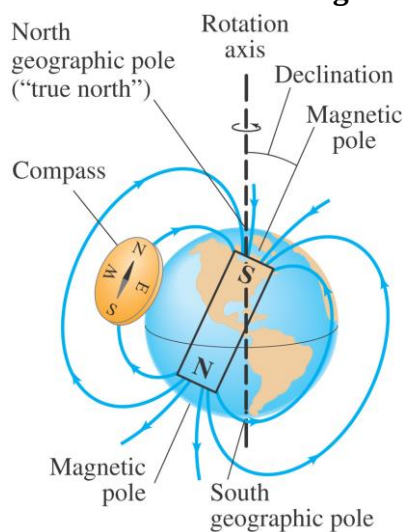
Magnetic Field Lines of Current Carrying Conductor (Straight Wire)



Magnetic Field Lines of Current Carrying Conductor (Circular Coil)



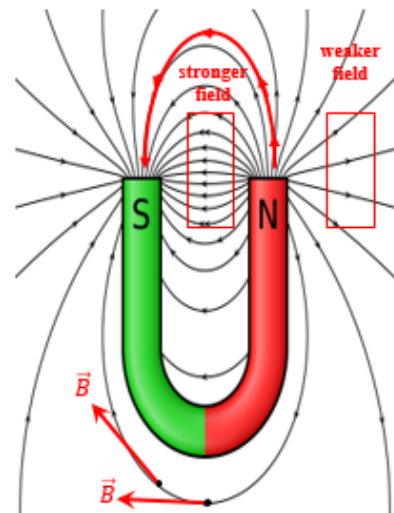
Magnetic Field Lines of Earth



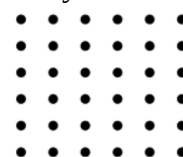
Note that the Earth's "North Pole" is really a south magnetic pole, as the north ends of magnets are attracted to it.

Sketch Magnetic Field Lines

1. Magnetic field lines **leave the north pole** and **enters south pole** of a magnet.
2. Can be represented by **straight lines** or **curves**.
3. The lines **do not intersect** one another.
4. The **direction of magnetic field vector $\vec{B}$** is **tangent to field line**.
5. The **number of lines per unit area** is **proportional** to the **strength** of the magnetic field.



6. Magnetic field lines enter **into** the page perpendicularly.
7. Magnetic field lines **out** of the page perpendicularly.



5.2 MAGNETIC FIELD PRODUCED BY CURRENT-CARRYING CONDUCTOR

Magnetic Field

When a **current flows in a conductor wire or coil**, the **magnetic field will be produced**.

The magnetic field produced at any point in the space is denoted as **B**.

It is a **vector** quantity.

The SI Unit is **Tesla (T)**.

The **direction of magnetic field** around the wire or coil can be determined by using the **right hand grip rule**.

Magnetic Field of a Long Straight Wire

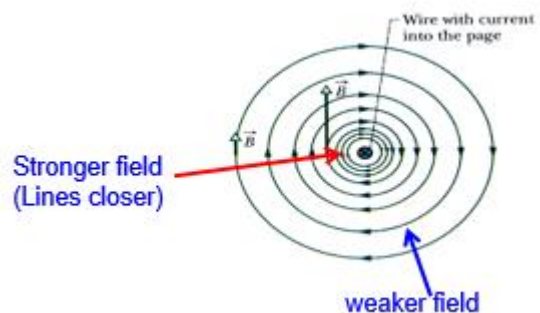
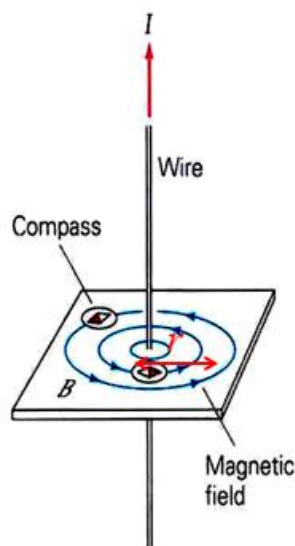
At a **perpendicular distance, r** from a long straight wire carrying a current I, the magnitude of **magnetic field, B** is given by:

$$B = \frac{\mu_0 I}{2\pi r}$$

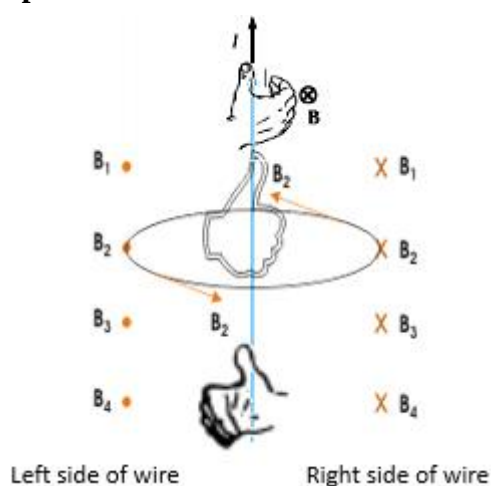
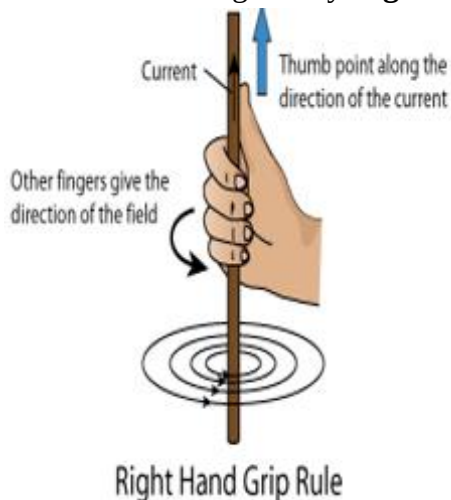
r: distance of any point from the straight conductor (wire)

I : current

$\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$ (permeability of free space)

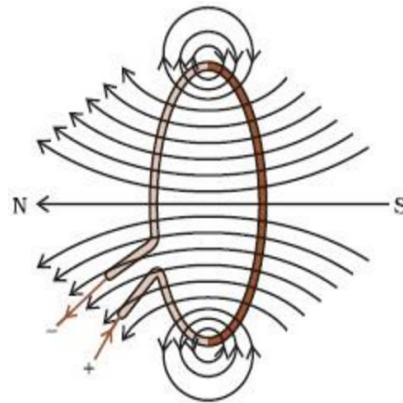
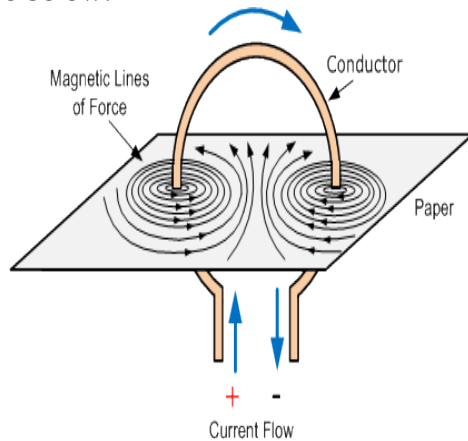


The direction of B is given by **Right Hand Grip Rule**



Magnetic Field at the Centre of a Circular Coil

Consider a **circular shaped conductor with radius r** that carries a current I as shown in figure below:



The **magnitude of magnetic field intensity B at point O** (centre of the circular coil or loop), is given by:

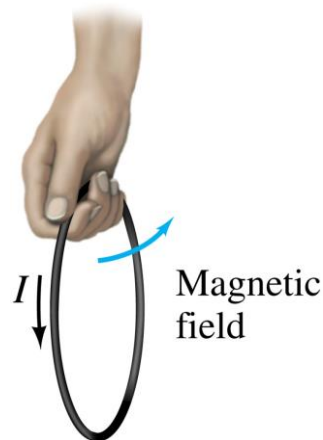
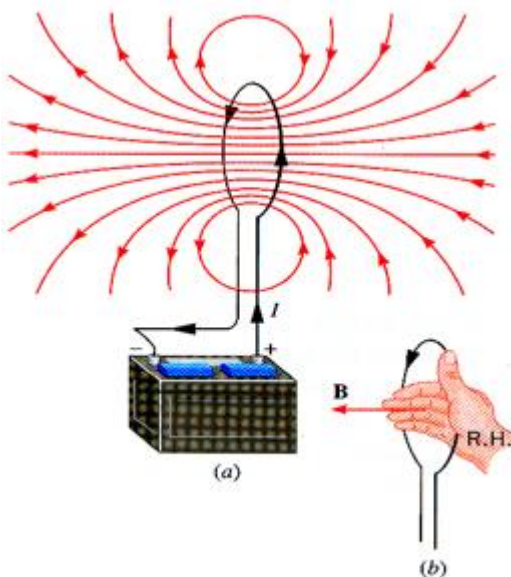
$$B = \frac{\mu_0 I}{2r}$$

r : radius of the circular coil

I : current

$\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$ (permeability of free space)

The direction of B is given by **Right Hand Grip Rule**



Thumb – **direction of current**
Other fingers – **direction of magnetic field (clockwise OR anticlockwise)**

For a circular coil consisting of **N turns, radius of the circular coil, r** and carrying current I , **magnetic field B** at the centre is given by:

$$B = \frac{\mu_0 N I}{2r}$$

r : distance of any point from the straight conductor (wire)

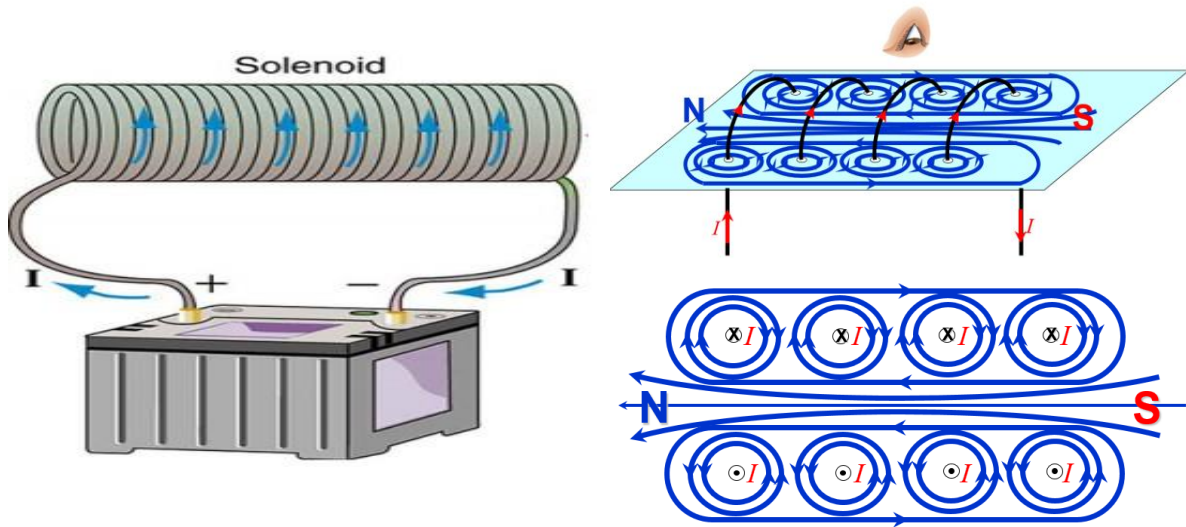
I : current

N : number of turns

$\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$ (permeability of free space)

Magnetic Field of a Solenoid

A solenoid is constructed by winding a long wire into a tight coil or helix with many circular loops.



The magnetic field is given by:

$$B = \mu_0 n I$$

I = current

$\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$ (permeability of free space)

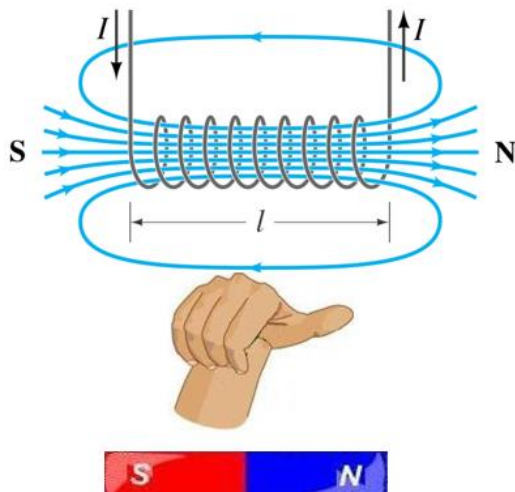
n = number of turns per unit length

$$n = \frac{N}{L}$$

N = total number of turns

L = length of the solenoid

The direction of B is given by **Right Hand Grip Rule**



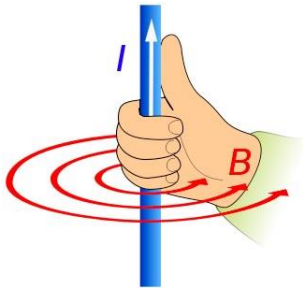
The directions of the fields can be found by viewing the current flows in the solenoid from both end or applying the right hand grip rule.

Thumb – **north pole**

Other fingers – **direction of current in solenoid.**

Direction of Magnetic Field using Right Hand Grip Rule

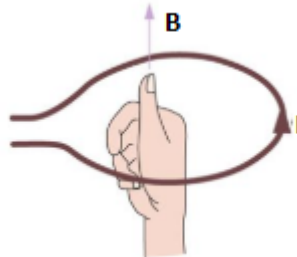
Long Straight Wire



Thumb:
Direction of **current**
Fingers:
Direction of **magnetic field**



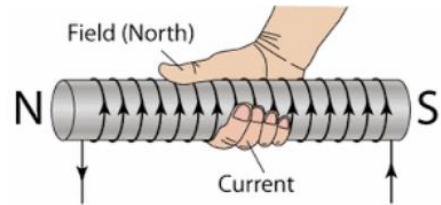
Circular Coil



Thumb:
Direction of **magnetic field**
Fingers:
Direction of **current**



Solenoid



Thumb:
Direction of **magnetic field**
Fingers:
Direction of **current**



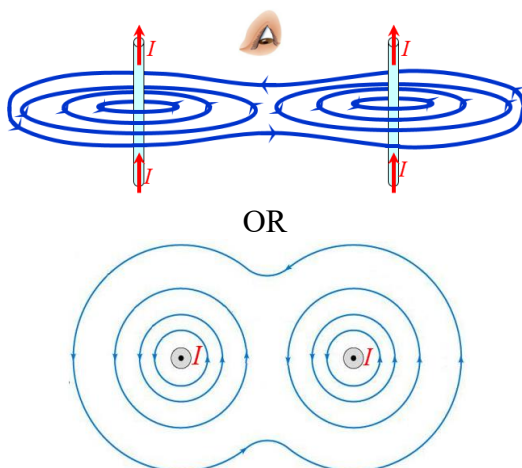
Example 1

Two long straight wires are placed parallel to each other and carrying the same current I . Sketch the magnetic field lines pattern around both wires

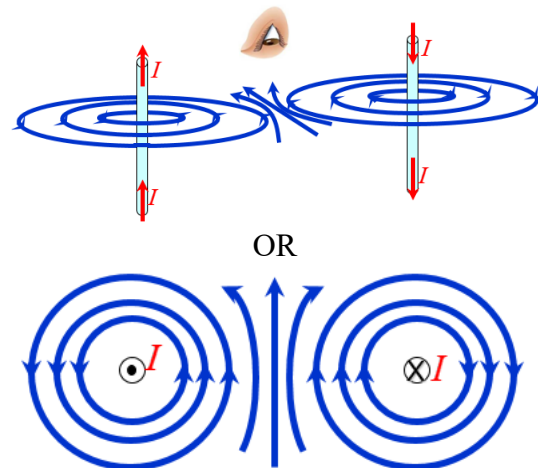
- When the currents are in the same direction.
- When the currents are in opposite direction.

Answers:

a)



b)



Example 2

A long wire (X) carrying a current of 50 A is placed parallel to and 5.0 cm away from a similar wire (Y) carrying a current of 10 A. Determine the magnitude and direction of the magnetic flux density at a point midway between the wires

- a) When the current are in the same direction.
b) When they are in opposite direction.

Answers:

$$I_X = 50 \text{ A}; I_Y = 10 \text{ A}; d = 5.0 \times 10^{-2} \text{ m} \rightarrow r_X = r_Y = 2.5 \times 10^{-2} \text{ m}$$

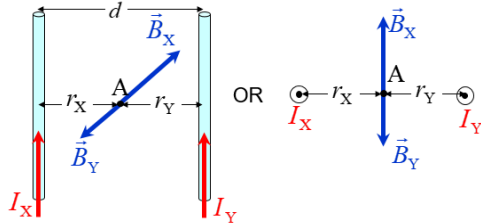
$$B_X = \frac{\mu_0 I_X}{2\pi r_X} = \frac{(4\pi \times 10^{-7})(50)}{2\pi(2.5 \times 10^{-2})}$$

$$B_X = 4.0 \times 10^{-4} \text{ T}$$

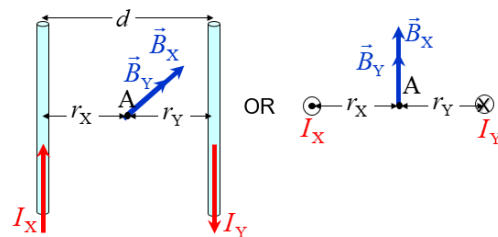
$$B_Y = \frac{\mu_0 I_Y}{2\pi r_Y} = \frac{(4\pi \times 10^{-7})(10)}{2\pi(2.5 \times 10^{-2})}$$

$$B_Y = 8.0 \times 10^{-5} \text{ T}$$

a) $B_X = -4.0 \times 10^{-4} \text{ T (into page)}$
 $B_Y = 8.0 \times 10^{-5} \text{ T out of page}$
 $B_{\text{net}} = -B_X + B_Y$
 $B_{\text{net}} = -4.0 \times 10^{-4} + 8.0 \times 10^{-5}$
 $B_{\text{net}} = -3.2 \times 10^{-4} \text{ T (into page)}$



b) $B_X = -4.0 \times 10^{-4} \text{ T (into page)}$
 $B_Y = -8.0 \times 10^{-5} \text{ T (into page)}$
 $B_{\text{net}} = -B_X + (-B_Y)$
 $B_{\text{net}} = -4.0 \times 10^{-4} + (-8.0 \times 10^{-5})$
 $B_{\text{net}} = -4.8 \times 10^{-4} \text{ T (into page)}$

**Example 3**

A circular coil has a diameter of 45.0 cm. If the magnetic field strength at the center of the coil is $5.3 \times 10^{-5} \text{ T}$, find the current flowing in the coil.

Answers:

$$B = 5.3 \times 10^{-5} \text{ T}; D = 45.0 \text{ cm} = 45.0 \times 10^{-2} \text{ m} \rightarrow r = 22.5 \times 10^{-2} \text{ m}$$

$$B = \frac{\mu_0 I}{2r}$$

$$5.3 \times 10^{-5} = \frac{(4\pi \times 10^{-7})I}{2(22.5 \times 10^{-2})} \rightarrow I = 19.0 \text{ A}$$

Example 4

A solenoid of length 1.5 m and 2.6 cm in diameter carries a current of 18 A. The magnetic field inside the solenoid is 2.26 mT. Calculate the number of turns of the solenoid.

Answers:

$$B = 2.26 \text{ mT} = 2.26 \times 10^{-3} \text{ T}; L = 1.5 \text{ m}; I = 18 \text{ A}$$

$$d = 2.6 \text{ cm} \rightarrow r = 1.3 \text{ cm} = 1.3 \times 10^{-2} \text{ m}$$

$$B = \frac{\mu_0 N I}{L}$$

$$2.26 \times 10^{-3} = \frac{(4\pi \times 10^{-7})N(18)}{1.5} \rightarrow N = 150 \text{ turns}$$

5.3 MAGNETIC FLUX

Magnetic Flux

1. Magnetic flux is defined as the **scalar product** between the **magnetic flux density, B** with the **vector of the surface area, A** .
2. Magnetic flux is a scalar quantity and its unit is **tesla meter square Tm^2 or Webber (Wb)**.

$$\phi = \vec{B} \cdot \vec{A} = BA \cos \theta$$

ϕ : Magnetic flux

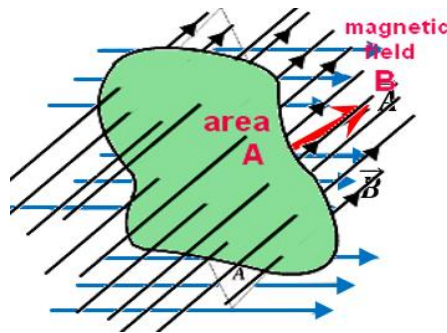
B : Magnitude of the magnetic flux density

A : Area at right angles to the magnetic field

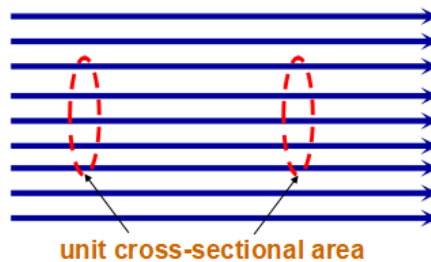
θ : angle between the direction of B and vector area A (normal to plane of the coil)

CAUTION : Direction of **vector $\vec{A}$** always **perpendicular to the surface area, A**

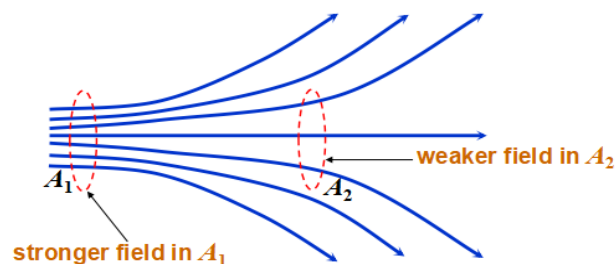
3. Magnetic flux is number of magnetic field lines that passing through a given area. The number of lines per unit cross-sectional area is proportional to the magnitude of the magnetic field. Magnetic field lines do not intersect one another.



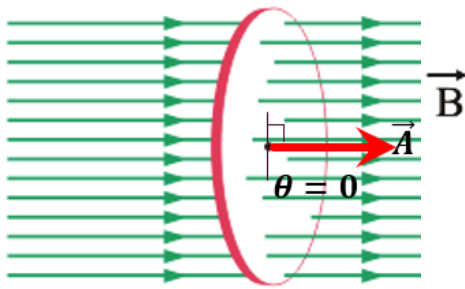
4. A uniform field is represented by parallel lines of force. This means that the number of lines passing perpendicularly through unit area at all cross-sections in a magnetic field are the same as shown in figure below:



5. A non-uniform field is represented by non-parallel lines. The number of magnetic field lines varies at different unit cross-sections as shown in figure below:



6. Let $A = 5 \text{ m}^2$ and $B = 6 \text{ T}$



Circular coil with magnetic field perpendicular to the plane of the coil.

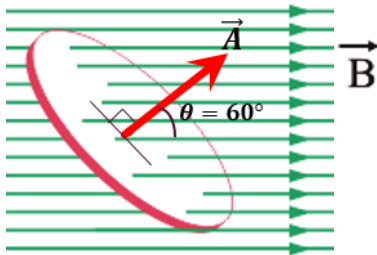
$$\theta = 0^\circ$$

$$\phi = BA \cos \theta$$

$$\phi = (6)(5) \cos 0$$

$$\phi = 30 \text{ Wb}$$

magnetic flux is maximum $\phi_{\max} = BA$



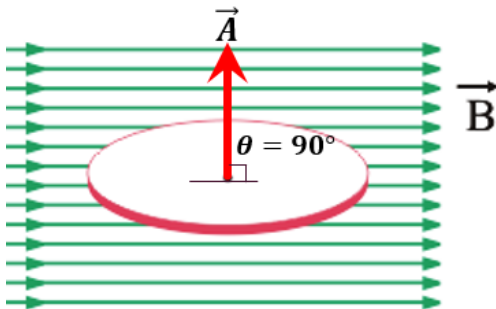
$$\theta = 60^\circ$$

$$\phi = BA \cos \theta$$

$$\phi = (6)(5) \cos 60$$

$$\phi = 15 \text{ Wb}$$

magnetic flux is $\phi > 0$



Circular coil with magnetic field parallel to the plane of the coil.

$$\theta = 90^\circ$$

$$\phi = BA \cos \theta$$

$$\phi = (6)(5) \cos 90$$

$$\phi = 0 \text{ Wb}$$

magnetic flux is $\phi = 0$

7. The flux can be visualized with respect to magnetic field lines. The value of the magnetic flux is proportional to the total number of lines passing through the loop.
8. When the area is perpendicular to the lines, the maximum number of lines passes through the area and the flux is a maximum.
9. When the area is parallel to the lines, no lines pass through the area and the flux is 0.

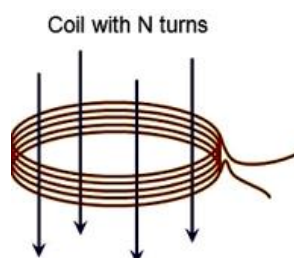
Magnetic Flux Linkage

Magnetic flux linkage (Φ) is a measure of the **total magnetic flux of passing through a coil with N turns.**

$$\Phi = N\phi$$

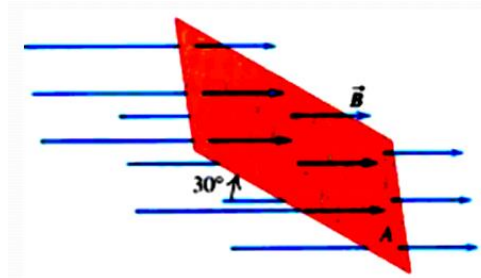
ϕ : Magnetic flux passing through one turn

N : Number of turns



Example 5

Figure shows a flat surface with an area of 3.0 cm^2 is placed in the uniform magnetic field. The plane surface makes an angle 30° with the direction of the magnetic field. If the magnetic flux through the surface is 0.90 mWb , calculate the magnitude of the magnetic field.



Answers:

$$\phi = 0.90 \text{ mWb} = 0.90 \times 10^{-3} \text{ Wb}$$

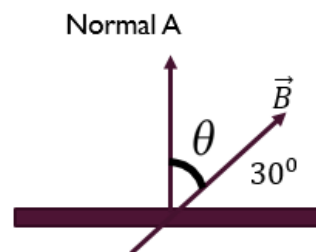
$$A = 3.0 \text{ cm}^2 = 3.0 \times 10^{-4} \text{ m}^2$$

$$\theta = 90^\circ - 30^\circ = 60^\circ$$

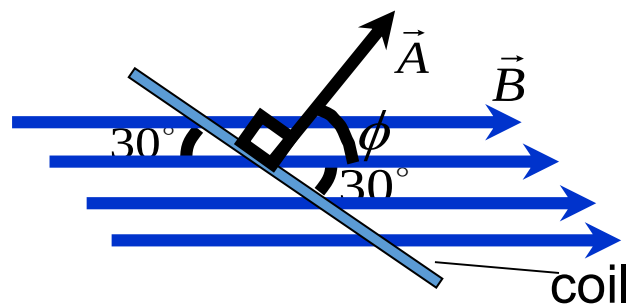
$$\phi = BA \cos \theta$$

$$0.90 \times 10^{-3} = B(3.0 \times 10^{-4})(\cos 60^\circ)$$

$$B = 6 \text{ T}$$

**Example 6**

A single turn of circular coil with a diameter of 3.0 cm is placed in the uniform magnetic field. The plane of the coil makes an angle 30° to the direction of the magnetic field. If the magnetic flux through the area of the coil is 1.20 mWb , calculate the magnitude of the magnetic field.



Answers:

$$\phi = 1.20 \text{ mWb} = 1.20 \times 10^{-3} \text{ Wb}$$

$$d = 3.0 \text{ cm} \rightarrow r = 1.5 \times 10^{-2} \text{ m}$$

$$A = \pi r^2 = \pi(1.5 \times 10^{-2})^2 = 7.07 \times 10^{-4} \text{ m}^2$$

$$\theta = 90^\circ - 30^\circ = 60^\circ$$

$$\phi = BA \cos \theta$$

$$1.20 \times 10^{-3} = B(7.07 \times 10^{-4})(\cos 60^\circ)$$

$$B = 3.40 \text{ T}$$

Example 7

A coil with 500 turns as a core with a radius of 2 cm. It is placed in a field of 0.6 T such that there is an angle of 30° between the field and the normal to the cross-sectional area. Calculate the magnetic flux and the magnetic flux linkage.

Answers:

$$B = 0.6 \text{ T}$$

$$r = 2 \text{ cm} = 2 \times 10^{-2} \text{ m}$$

$$A = \pi r^2 = \pi(2 \times 10^{-2})^2 = 1.26 \times 10^{-3} \text{ m}^2$$

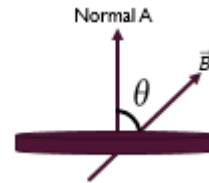
$$\theta = 30^\circ$$

Magnetic flux,

$$\phi = BA \cos \theta$$

$$\phi = (0.6)(1.26 \times 10^{-3}) \cos 30^\circ$$

$$\phi = 6.55 \times 10^{-4} \text{ Wb}$$



Magnetic flux linkage,

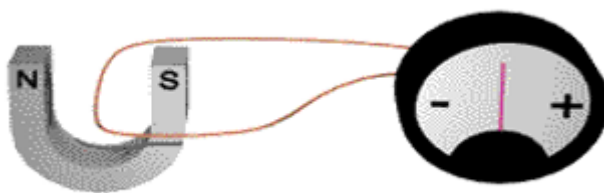
$$\Phi = N\phi$$

$$\Phi = (500)(6.55 \times 10^{-4})$$

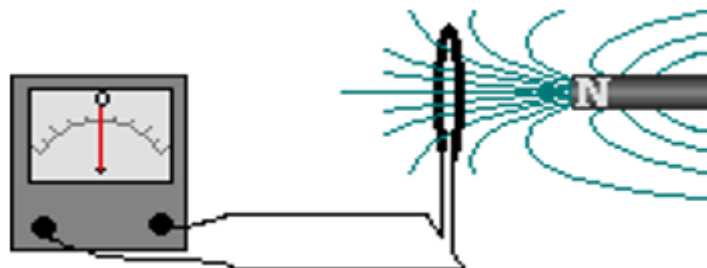
$$\Phi = 0.33 \text{ Wb}$$

5.4 INDUCED EMF

- There are two ways in which electricity and magnetism is related:
 - Electric current produces a magnetic field.
 - Magnetic field exerts a force on electric current and moving charge.
- Electromagnetic Induction** is the **production of induced emf** (or induced current) whenever the **magnetic flux** through a loop, coil or circuit is **changed**.
- The meaning of changing in magnetic flux:
 - There is a relative motion of loop & magnet fieldlines are 'cut'



- The number of magnetic field lines passing through a coil are increased or decreased.



Faraday's Law

1. **Faraday's law** of electromagnetic induction states that **the magnitude of the induced emf is proportional to the rate of change of the magnetic flux.**

$$\varepsilon = - \frac{d\phi}{dt}$$

ε : Induced emf

$d\phi$: change of magnetic flux

dt : change of time

2. The negative sign indicates that the direction of induced emf always oppose the change of magnetic flux producing it (Lenz's Law).
3. For a coil of N turns,

$$\varepsilon = -N \frac{d\phi}{dt}$$

or

$$\varepsilon = -N \frac{(\phi_f - \phi_i)}{dt}$$

Example 8

The magnetic flux passing through a single turn of a coil is increased quickly but steadily at a rate of $5.0 \times 10^{-2} \text{ Wb s}^{-1}$. If the coil has 500 turns, calculate the magnitude of the induced emf in the coil.

Answer:

$N = 500 \text{ turns};$

$$\frac{d\phi}{dt} = 5.0 \times 10^{-2} \text{ Wb s}^{-1}$$

By applying the Faraday's law equation for a coil of N turns, thus

$$\varepsilon = N \frac{d\phi}{dt} = (500)(5.0 \times 10^{-2})$$

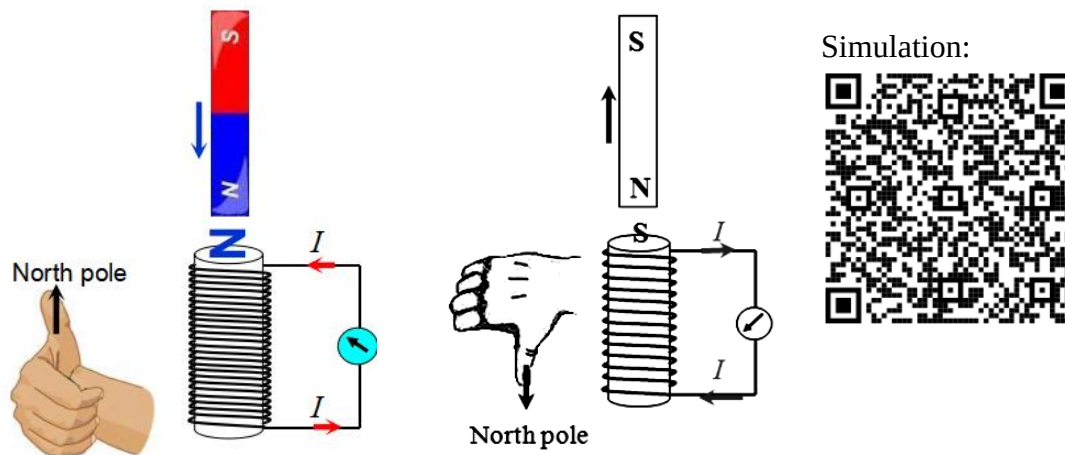
$$\varepsilon = 25 \text{ V}$$

Lenz's Law

1. **Lenz's Law** states that **an induced electric current always flows in such a direction that it opposes the change producing it.**
2. The direction of any magnetic induction effect is such as to oppose the cause of the effect.
3. The direction of an induced current or emf is such that the magnetic field created by induced current opposes the change in magnetic flux that created the current.

When applying Lenz's law,

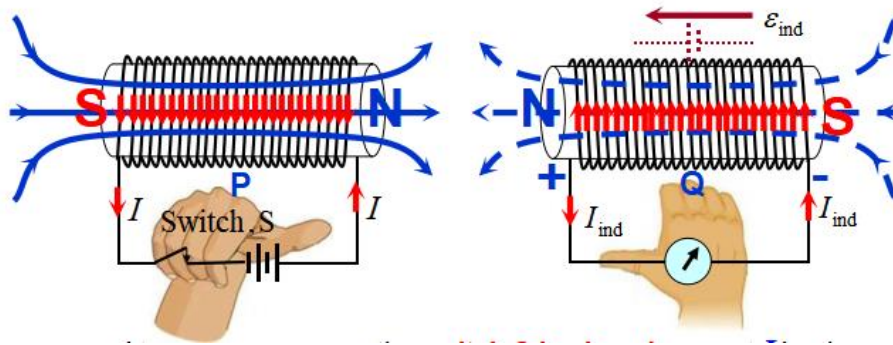
- Identify the change that causes the change in magnetic flux
- Direction of induced current must produce an effect that opposes the change

1st Experiment : Bar magnet is moved towards and away from solenoid.**When N-pole of the magnet is pushed into the solenoid**

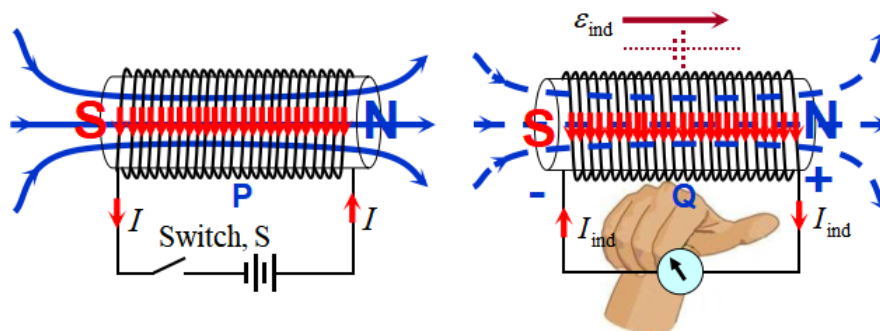
1. The magnitude of the magnetic field at the solenoid increases as the bar magnet is moved towards it.
2. According to Faraday's law, the increase in magnetic flux linkage induces an emf in the solenoid.
3. According to Lenz's law, induced current flows in a direction that would produce a magnetic field to oppose N-pole entering the solenoid.
4. The induced current in the solenoid produce an N-pole at the top end of the solenoid to repel the approaching N-pole of the magnet.
5. The direction of induced current can be determined by using right hand grip rule.

When the N-pole of the magnet is pulled out of the solenoid

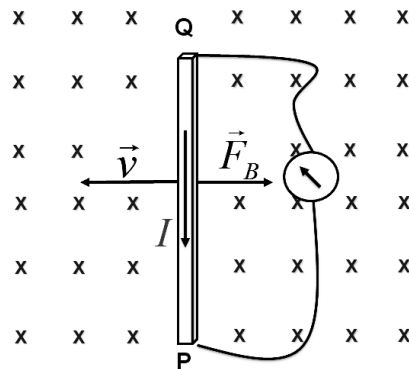
1. Magnetic flux linked with the solenoid decrease.
2. According to Faraday's law, the decrease in the magnetic flux linkage induces an emf in the solenoid.
3. According to Lenz's law, the induced current flows in a direction that would produce a magnetic field to attract the outgoing N-pole of the magnet.
4. The induced current in the solenoid produce an S-pole at the top end of the solenoid to attract the outgoing N-pole of the magnet.
5. The direction of induced current can be determined by using right hand grip rule.

2nd Experiment : Two solenoids arranged coaxially close to each other**Switch S closed**

- 1) At the moment when the switch S is closed, current I begins to flow in the solenoid P and produces a magnetic field inside the solenoid P. Suppose that the field points towards the solenoid Q.
- 2) The magnetic flux through the solenoid Q increases with time. According to Faraday's law, an induced current due to induced emf will exist in solenoid Q.
- 3) According to Lenz's law, the induced current flows in solenoid Q must produce a magnetic field that opposes the N-pole of solenoid P. The induced current in solenoid Q produces an N-pole at the left end of the solenoid to repel the N-pole of solenoid P.
- 4) The direction of induced current in wire Q can be determined by using right hand grip rule.

Switch S opened

- 1) At the moment when the switch S is opened, the current starts to decrease in the solenoid P and magnetic flux through the solenoid Q decreases with time.
- 2) According to Faraday's law, an induced current due to induced emf will exist in solenoid Q.
- 3) According to Lenz's law, the induced current flows in solenoid Q must produce a magnetic field that opposes the change producing it (decrease in flux).
- 4) Therefore, the induced current in solenoid Q produces an S-pole at the left end of the solenoid.
- 5) The direction of induced current can be determined by using right hand grip rule.

3rd Experiment : Straight conductor is placed perpendicular to the magnetic field.**A straight conductor in a uniform magnetic field**

- 1) When the conductor moves with constant velocity, v , to the left, the change in magnetic flux induces an emf in the conductor.
- 2) According to Lenz's law, induced current flows in a direction so that it will produce a force that opposes the motion of the conductor.
- 3) Therefore, the magnetic force on the conductor is to the right.
- 4) The direction of current flow in the conductor can be determined by using **right hand rule**. Therefore, the induced current flows from Q to P.

Example 9

A bar magnet is moved rapidly toward a 40 turn circular coil of wire. As the magnet moves, the average value of $B \cos \theta$ over the area of the coil increase from 0.0125 T to 0.45 T in 0.250 s. If the radius of the coils is 3.05 cm, and the resistance of its wire is 3.55 Ω , Find

- a) The magnitude of the induced emf.
- b) The magnitude of the induced current.

Answers:

$$N = 40 \text{ turns}; t_f - t_i = 0.25 \text{ s}; R = 3.55 \Omega$$

$$B_i \cos \theta = 0.0125 \text{ T}; B_f \cos \theta = 0.45 \text{ T}$$

$$r = 3.05 \text{ cm} = 3.05 \times 10^{-2} \text{ m} \rightarrow A = \pi r^2 = \pi(3.05 \times 10^{-2})^2 = 2.92 \times 10^{-3} \text{ m}^2$$

a)

$$\phi_i = (B_i \cos \theta)A = (0.0125)(2.92 \times 10^{-3}) = 3.65 \times 10^{-5} \text{ Tm}^2$$

$$\phi_f = (B_f \cos \theta)A = (0.45)(2.92 \times 10^{-3}) = 1.32 \times 10^{-3} \text{ Tm}^2$$

$$\varepsilon = -N \frac{(\phi_f - \phi_i)}{t_f - t_i}$$

$$\varepsilon = -(40) \frac{(1.32 \times 10^{-3} - 3.65 \times 10^{-5})}{0.25}$$

$$\varepsilon = 0.205 \text{ V}$$

b)

Using Ohm's law,

$$I = \frac{V}{R} = \frac{0.205}{3.55}$$

$$I = 0.0577 \text{ A}$$

Example 10

The induced emf in a single loop of wire has a magnitude of 1.48 V when the magnetic flux is changed from 0.85 Tm^2 to 0.11 Tm^2 . How much time is required for this change flux?

Answer:

$$N = 1 \text{ turn}; \phi_i = 0.85 \text{ Tm}^2; \phi_f = 0.11 \text{ Tm}^2; \varepsilon = 1.48 \text{ V}$$

$$\varepsilon = -N \frac{(\phi_f - \phi_i)}{t_f - t_i} \rightarrow 1.48 = -(1) \frac{(0.11 - 0.85)}{t_f - t_i}$$

$$t_f - t_i = 0.5 \text{ s}$$

Example 11

A single turn circular shaped coil has resistance of 100Ω and It moves towards the north pole of a bar magnet as shown in figure below. If the average rate of change of magnetic flux through the coil is $2.5 \times 10^{-4} \text{ Wbs}^{-1}$, determine the induced current in the coil and state the direction of the induced current observed by the observer shown in figure.

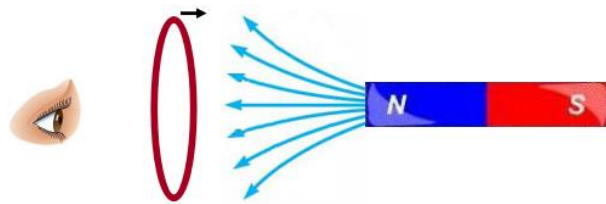
Answers:

$$\varepsilon = 2.5 \times 10^{-4} \text{ V}; R = 100 \Omega$$

By using Ohm's law,

$$I = \frac{V}{R} = \frac{2.5 \times 10^{-4}}{100}$$

$$I = 2.5 \times 10^{-6} \text{ A}$$



Direction of the induced current observed by the observer clockwise.

Example 12

A bar magnet is held above a loop of wire in a horizontal plane, as shown in figure.

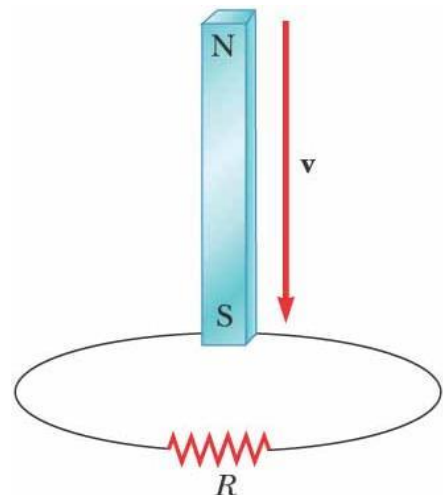
The south end of the magnet is toward the loop of the wire. The magnet is dropped toward the loop. From top view, find the direction of the current through the resistor

- While the magnet falling toward the loop.
- After the magnet has passed through the loop and moves away from it.

Answers:

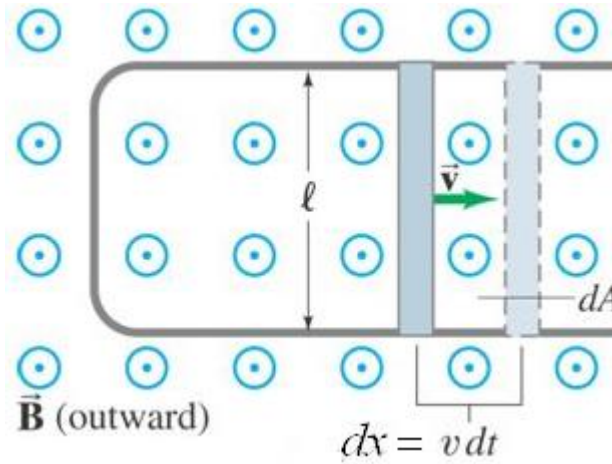
a) Clockwise

b) Counterclockwise



Induced emf in a Straight Conductor Moving in a Magnetic Field

Consider a linear (straight) conductor of length l is moved with velocity v across a uniform magnetic field B as shown in figure below.



When the conductor moved through a distance x in time t , the area swept out by the conductor is given by

$$A = lx$$

Since the motion of the conductor is perpendicular to the magnetic field B hence the magnetic flux cut by the conductor is given by

$$\phi = BA \cos 0^\circ, \phi = Blx$$

According to Faraday's law (noting that $N = 1$), the emf is induced in the conductor and its magnitude is given by

$$\varepsilon = \frac{d\phi}{dt} = \frac{d(Blx)}{dt} = Bl \frac{dx}{dt}, \quad \text{but } \frac{dx}{dt} = v$$

Therefore, $\varepsilon = Blv$

In general,

$$\varepsilon = Blv \sin \theta$$

θ : angle between $\vec{v}$ and $\vec{B}$

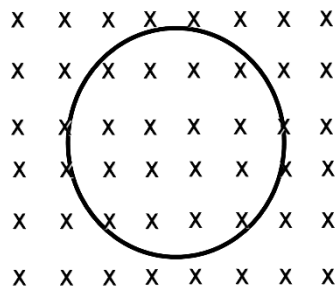
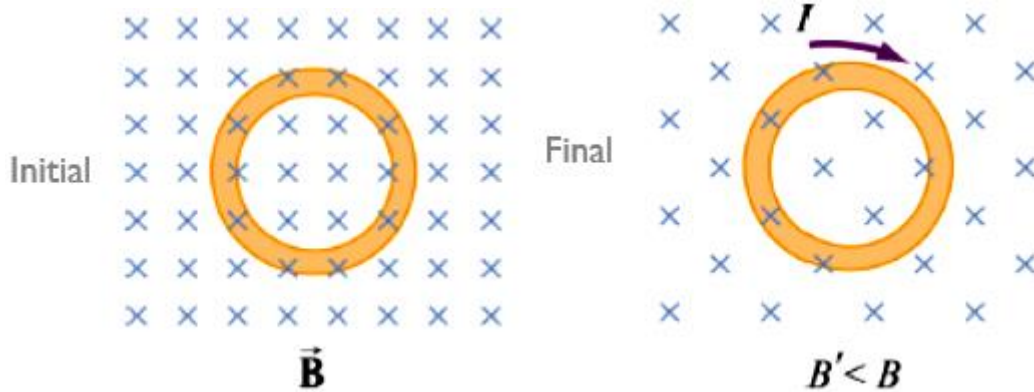
Induced emf In Coil

Figure shows a coil of N turns each of area A in a magnetic flux density B .

The magnetic flux linkage with the coil is given by $\Phi = NBA$.

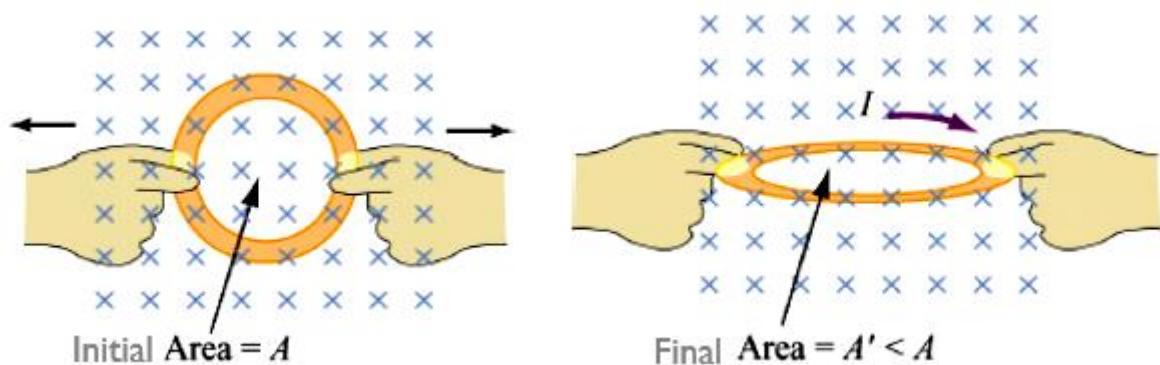
According to Faraday's law, when the magnetic flux linkage changes, an emf is induced in the coil

$$\varepsilon = -\frac{d\Phi}{dt} \quad , \quad \varepsilon = -\frac{d(NBA)}{dt}$$

Induced emf in Coil by Changing Magnetic Field Strength, B with Time

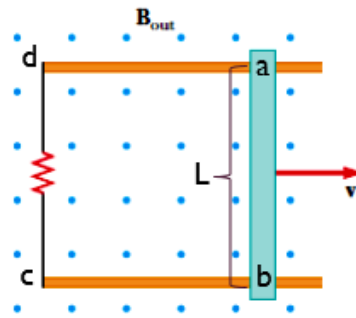
If $\vec{B}$ is perpendicular to the plane of coil $\rightarrow \theta = 0^\circ$, $\cos 0^\circ = 1$ & area, A is constant, then

$$\varepsilon = -NA \frac{dB}{dt}$$

Induced emf in Coil by Changing the Area, A of the Coil with Time

If $\vec{B}$ is perpendicular to the plane of coil $\rightarrow \theta = 0^\circ$, $\cos 0^\circ = 1$ & B is constant, then

$$\varepsilon = -NB \frac{dA}{dt}$$

Example 13

Assume that $R = 12\ \Omega$, $L = 1.2\ \text{m}$ & a uniform $2.5\ \text{T}$ magnetic field is directed out the page.

- At what speed should the bar be moved to produce a current of $0.5\ \text{A}$ in the resistor.
- What is the direction of the induced current?

Answers:

$$R = 12\ \Omega; l = 1.2\ \text{m}; B = 2.5\ \text{T}; \theta = 90^\circ; I = 0.5\ \text{A}$$

a)

$$\varepsilon = IR = (0.5)(12) = 6\ \text{V}$$

$$\varepsilon = Blv \sin \theta$$

$$6 = (2.5)(1.2)v \sin 90^\circ$$

$$v = 2.0\ \text{m s}^{-1}$$

b)

Clockwise (abcd)

Example 14

A coil having an area of $8.0\ \text{cm}^2$ and 50 turns lies perpendicular to a magnetic field of $1.8\ \text{T}$. If the magnetic flux density is steadily reduced to $0.6\ \text{T}$ in $0.02\ \text{s}$, determine the induced emf.

Answer:

$$A = 8.0\ \text{cm}^2 = 8.0 \times 10^{-4}\ \text{m}^2; N = 50\ \text{turns}; B_i = 1.8\ \text{T}; B_f = 0.6\ \text{T}; dt = 0.02\ \text{s}$$

$$dB = B_f - B_i = 0.6 - 1.8 = -1.2\ \text{T}$$

induced emf in coil by changing magnetic field strength, B with time,

$$\varepsilon = -NA \frac{dB}{dt} = -(50)(8.0 \times 10^{-4}) \frac{(-1.2)}{0.02} = 2.4\ \text{V}$$

Example 15

A narrow coil of 10 turns and area of $48\ \text{m}^2$ is placed perpendicular to a uniform magnetic field of $1.2\ \text{T}$. After 5 s, the area of the coil is increased to $83\ \text{m}^2$. Calculate the emf induced.

Answer:

$$B = 1.2\ \text{T}; N = 10\ \text{turns}; A_i = 48\ \text{m}^2; A_f = 83\ \text{m}^2; dt = 5\ \text{s}$$

$$dA = A_f - A_i = 83 - 48 = 35\ \text{m}^2$$

induced emf in coil by changing the area, A of the coil with time

$$\varepsilon = -NB \frac{dA}{dt} = -(10)(1.2) \frac{(35)}{5} = -84\ \text{V} \text{ (-ve Lenz's law)}$$

Chapter 5 MCQs Practice:



SELECTED FORMULAE

1. $B = \frac{\mu_0 I}{2\pi r}$

2. $B = \frac{\mu_0 I}{2r}$

3. $B = \mu_0 nI$

4. $\phi = BA \cos \theta$

5. $\Phi = N\phi$

6. $\varepsilon = -\frac{d\phi}{dt}$

7. $\varepsilon = Blv \sin \theta$

8. $\varepsilon = -NA \frac{dB}{dt}$

9. $\varepsilon = -NB \frac{dA}{dt}$

DIY Quiz

Instruction:

Answer all questions in 10 minutes.

No discussion & do not refer to any notes.

You can only refer to the formulae given below.

1. A 30 cm long solenoid carries a current of 2 A. If the magnetic field at the center of the solenoid is 4×10^{-3} T, calculate the number of turns of the solenoid.
2. A circular loop of radius 8 cm consists of 20 turns placed in a uniform magnetic field of 0.1 T. The direction of the magnetic field makes an angle of 38° with the plane of the loop. Calculate the magnetic flux linkage.
3. A coil of 300 turn and area $2 \times 10^{-4} \text{ m}^2$ is placed in uniform magnetic field which is changing from 0 T to 3 mT in 2 s. Calculate the maximum emf generated.

Answers:

1. 477 turns
2. 0.25 Wb
3. $9 \times 10^{-5} \text{ V}$

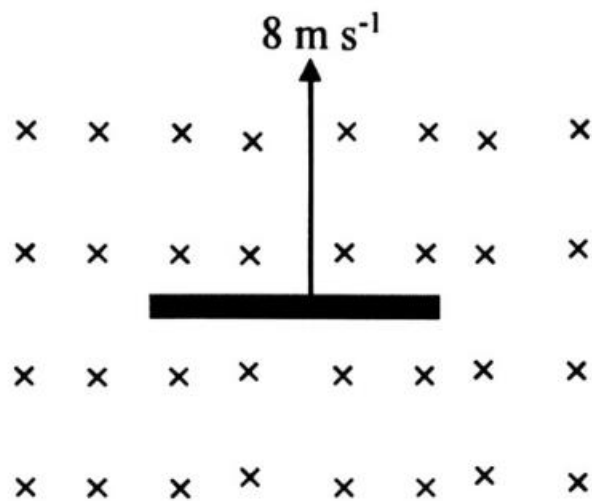
PAST YEAR QUESTIONS

Figure above shows a 1.6 m long copper rod with resistance 5Ω being pulled upward at a constant speed of 8 m s^{-1} perpendicular to a uniform 3.2 mT magnetic field.

- Calculate induced emf in the copper rod.
- Calculate magnitude of induced current through the copper rod.
- State direction of induced current through the copper rod.

[6 marks]